Eaton 9130 Series

DSP firmware upgrade instructions

10 28 349 Revision P4
2012-04-27

Introduction

This document gives a guidance of how to upgrade (flash) the Main (DSP) of Powerware 9130 series UPS. Available firmware updates and required software tools can be downloaded from http://www.powerware.com/software/downloads.asp.

Purpose

Periodically, Eaton releases firmware upgrades for some of its products. With each new release, new features become available and often, some issues are improved. Please reference the document *Tracking of Modify table V1.28.doc* in the firmware upgrade package if you are interested in knowing what enhancements and changes have been made between each released version of firmware.

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Quick reference

Main (DSP) firmware upgrade via RS-232

1. Shutdown the devices supported by the UPS.
2. Ensure the UPS is in Standby state (output off).
3. Set communication speed of RS-232 port (and USB) in the UPS to 9600bps: *Settings → User Settings → Serial Port Configuration → Port: RS232, Speed: 9600.*
4. Connect RS-232 cable between the computer and the UPS.
5. Start ***Watcher - UpsFlash*** software.
6. Select COM port and Bit Rate 9600bps and press “Open”
7. Press “Open File”, and select an “S” file to be flashed (e.g. *V1.22.S*).
8. Press “Flash Normal” and unplug input power cord from the UPS.
9. Wait until “Flash ok” notice will appear and press “Close” button.

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1 Main (DSP) firmware upgrading

1.1 Requirements

In order to upgrade the main (DSP) firmware of the UPS, you will need:

- a computer with Microsoft Windows operating system (Windows 2000 or newer)
- an RS-232 or USB cable (both are included in the package with the UPS)
- *Watcher – UpsFlash* software package (can be downloaded from http://www.powerware.com/software/downloads.asp)

The software package (.zip file) includes the files:

- *Watcher – UpsFlash Vx.x.exe*, the DSP firmware upgrade tool
- *UPSDeviceList.ini*
- *Vx.xx.S*, the firmware upgrade (e.g. *V1.22.S*)
- *Tracking of Modify table Vx.xx.doc,* list of changes (optional)
- *1028349 rev. X…,* this guide.

Unpack or Extract the package to a temporary directory.

1.2 DSP Firmware Upgrading via RS-232 – Flash Normal

Follow the procedure below while upgrading the DSP firmware of the UPS via RS-232 cable.

1. If the UPS is feeding any load (computer or other equipment), turn off the load(s) supported by the UPS. Then command the UPS to standby state by pushing ⏻ button for at least three seconds.

   In case the UPS is off completely, provide AC supply by plugging the input power cord to the unit when it should automatically enter to standby state.

   State of the UPS can be verified from the front panel's *UPS Status* menu:

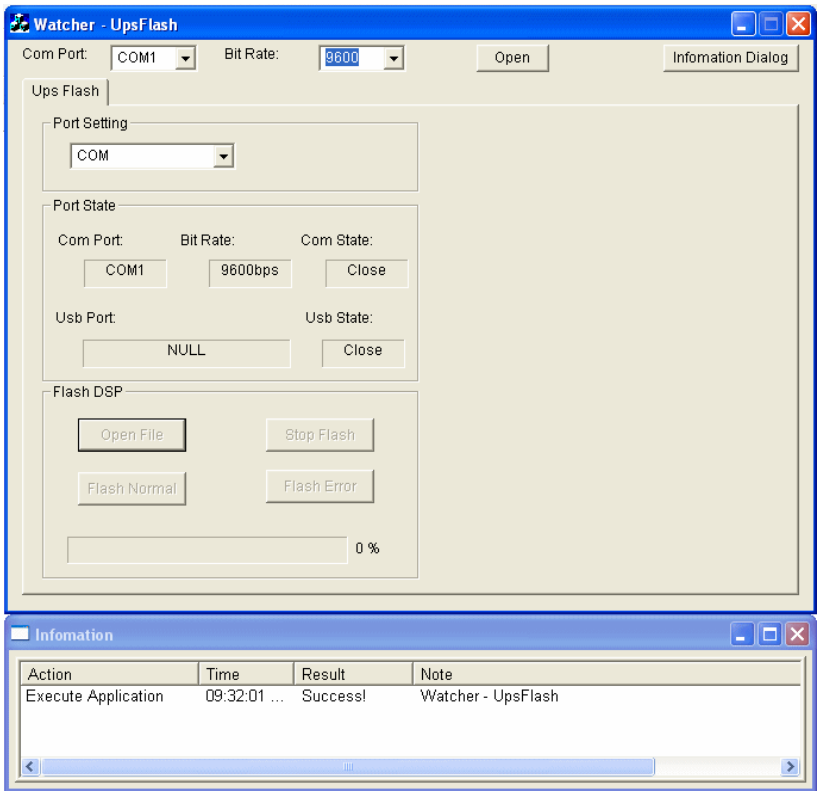
2. Set communication speed of the RS-232 port (and USB) in the UPS to 9600 bps using the LCD front panel: Go to *Settings → User Settings → Serial Port Configuration → Port: RS232, Speed: 9600.*

3. Connect the RS-232 cable between DB-9 connectors of the computer and the UPS.

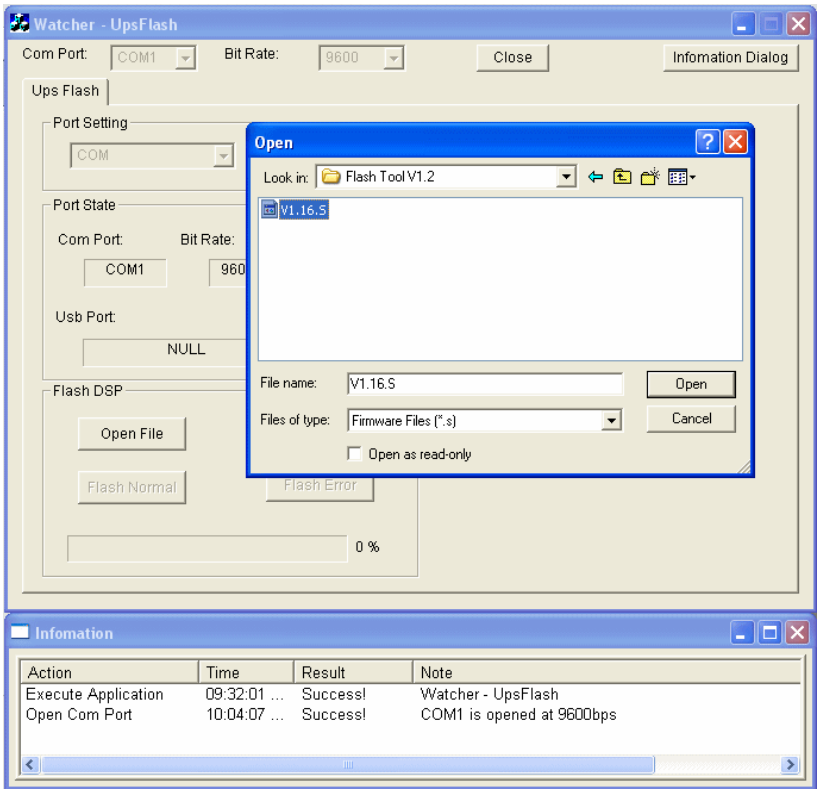
4. Start ***Watcher - UpsFlash*** software from the computer.

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5. Choose the right COM port (typically COM1) and Bit rate “9600”, then press Open to open COM port. Choose “**Ups Flash**” Tab.



6. Press Open File and select “S” file to be flashed and click Open.



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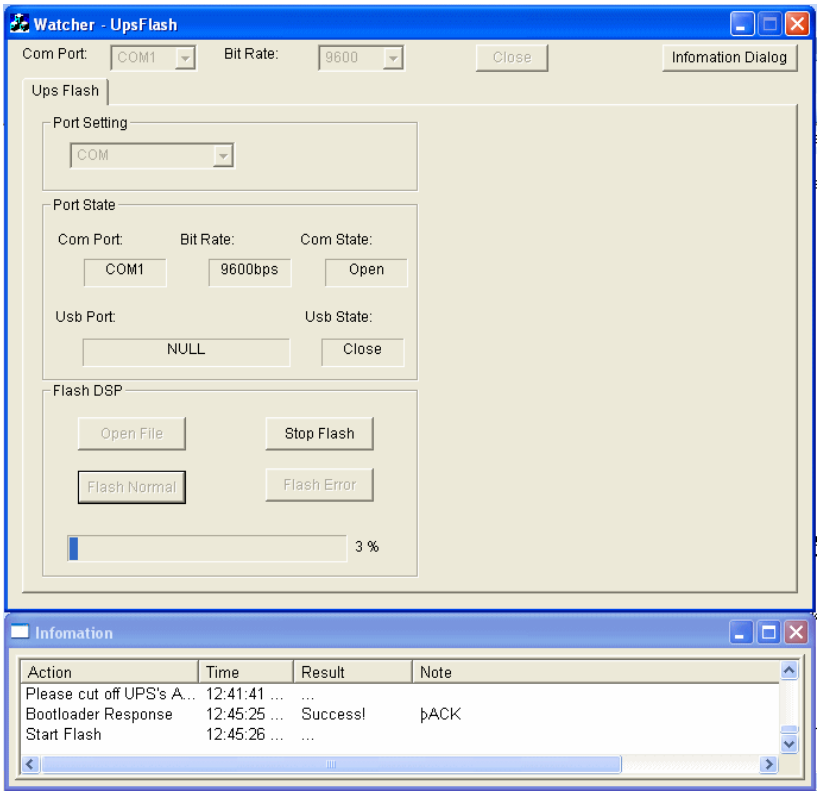
7. Press Flash Normal button when the following Popup Message Box should appear:



Note. If the message below will appear instead, return to item 1 in this procedure to verify the cabling, COM port settings and that the UPS is in standby state.



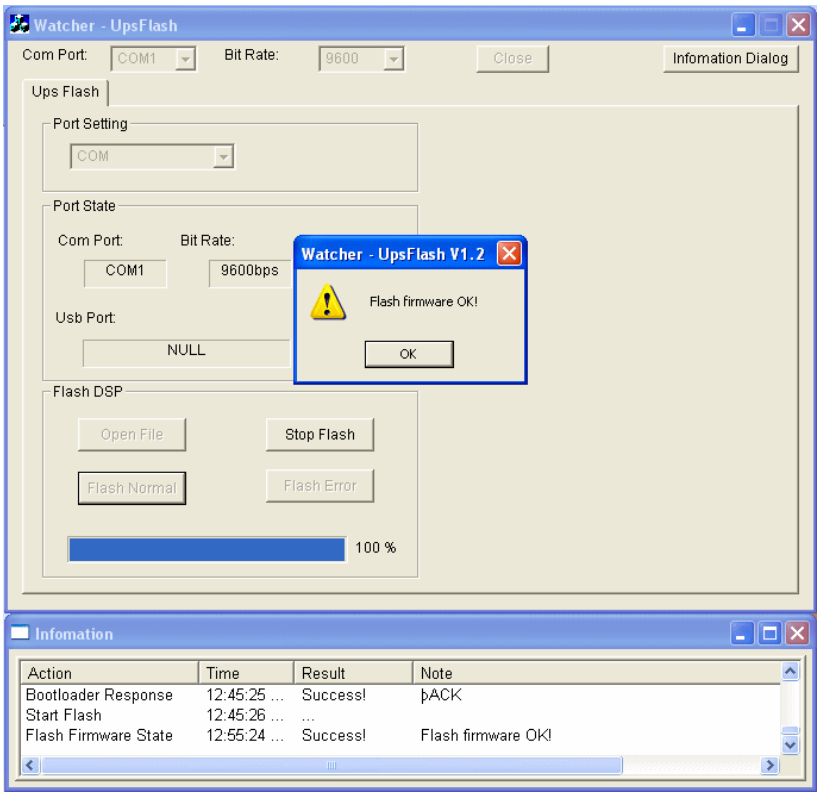
8. Unplug the input power cord from the UPS and click OK. The flashing will begin after a few seconds.



Note. If "Flash firmware NG!" message will appear during the flashing, go to Chapter *1.3 DSP Firmware Upgrading via RS-232 – Flash Error*.

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9. The LCD display “Flashing in progress…”. Wait until the “Flash OK” message will appear. Click OK and Close buttons and exit from the software.

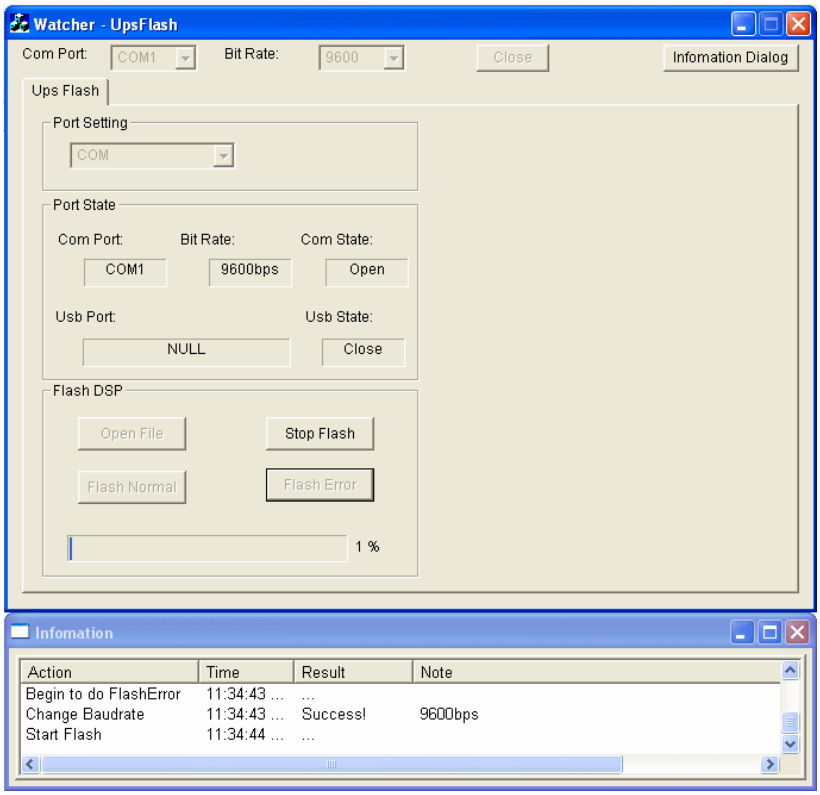


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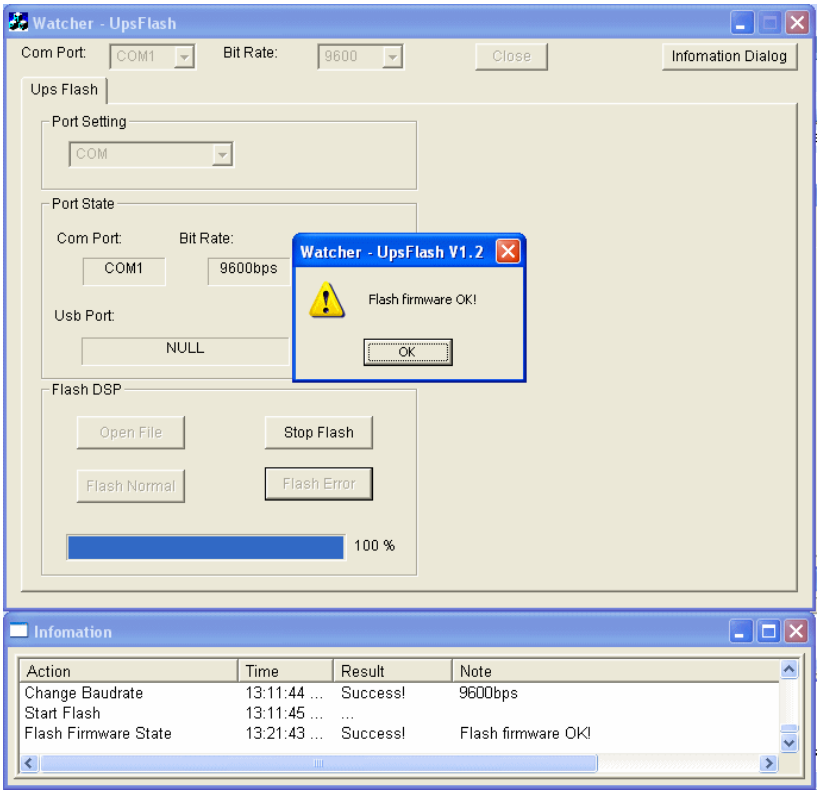
1.3 DSP Firmware Upgrading via RS-232 – Flash Error

This procedure is used when the DSP firmware upgrading via RS-232 port has failed or interrupted with “Flash firmware NG!” message (buzzer of the UPS may beep all the time).

1. Keep the input power cord disconnected from the UPS.
2. If ***Watcher - UpsFlash*** software is not running, start up the software and choose the right COM port (typically COM1) and Bit rate “9600”, then press Open to open COM port.
3. Press Open File button and select “S” file to be flashed (e.g. *V1.22.S*) and click Open.
4. Click Flash Error button. The DSP flashing should begin after a few seconds.

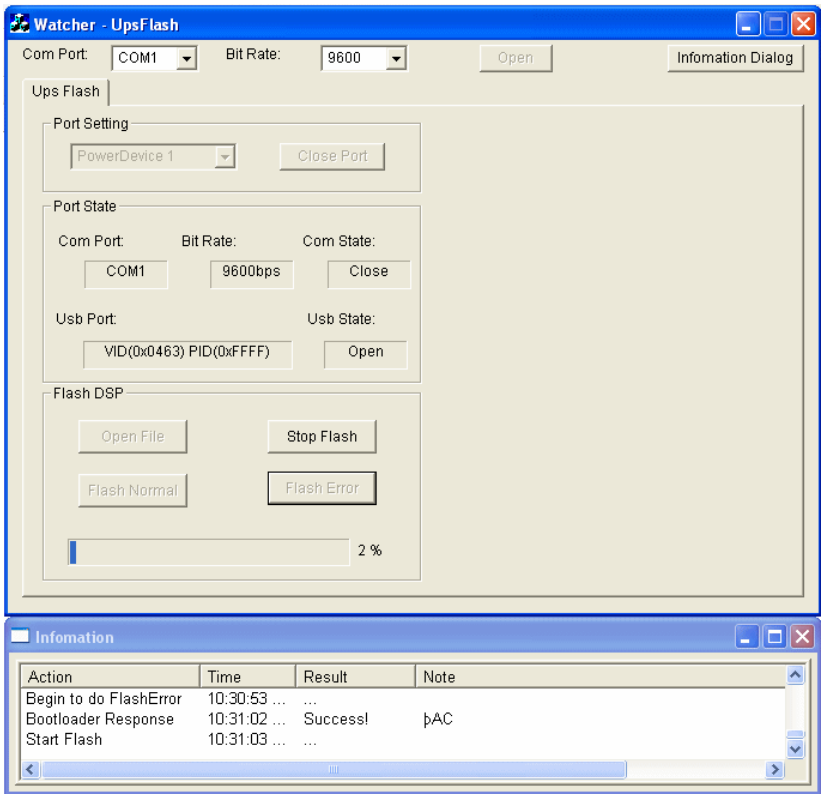


5. Wait until the “Flash OK” message will appear. Click OK and Close buttons and exit from the software.



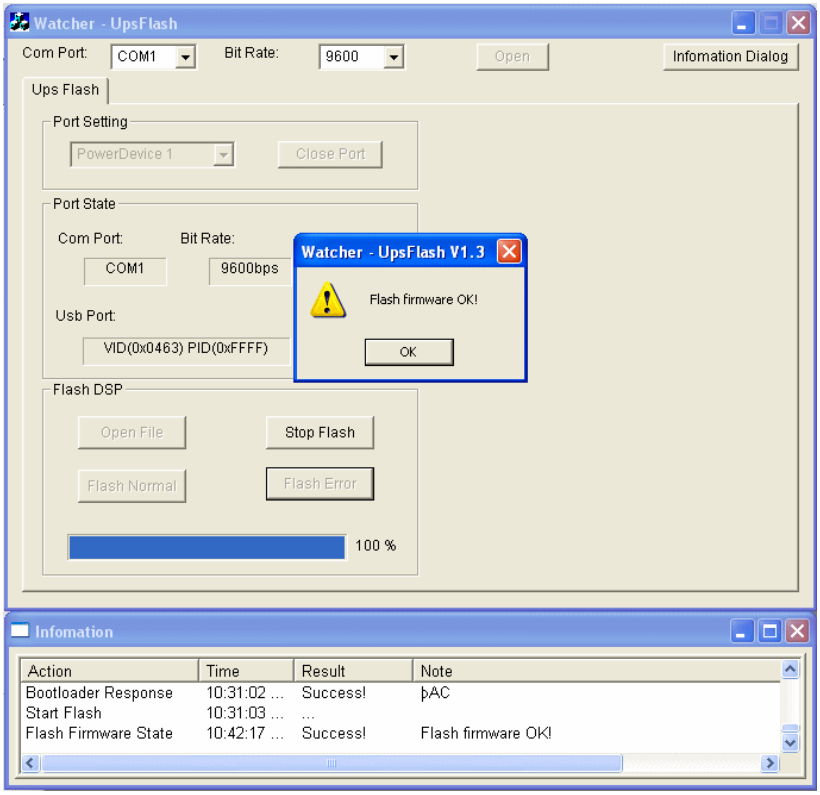
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1. Wait until the “Flash OK” message will appear. Click OK and Close buttons and exit from the software.



2. Restart LanSafe Power Monitor service if it was stopped by running *Services.msc* from Windows command prompt and there click on LanSafe Power Monitor and select “Start the service”.

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